

Gene Pulser® Electroporation

Cell Type Bacterial, gram positive

Molecules Electroporated DNA: pLAR 33 ,18 kB plasmid, double stranded DNA .

Species Used *Lactobacillus* sp., strain ES1

Before the Pulse

Cell Growth Medium MRS broth (DIFCO)

Growth Phase at Harvest O.D. (600) =1.0 (log)

Pre-pulse Incubation Buffer, cuvettes, cells placed on ice

Wash Solution 2% glycerol in 7 mM NaPO₄ (pH 6.0)

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 25 °C

Electroporation Medium 2% glycerol in 10 mM NaPO₄ (pH 6.0)

Cuvette Gap 0.2 cm

Cell Density Cells concentrated 80X

Voltage 2.0 kV

Volume of Cells 50 µl

Field Strength 10 kV/cm

DNA Concentration 1 µg

DNA Resuspension Buffer TE buffer (10 mM Tris, 1 mM EDTA,pH 8.0)

Capacitor 25 µF

Volume of DNA 2 µl

Resistor (Pulse Controller) 200 Ω

After the Pulse

Time Constant 2.2 msec

Outgrowth Medium MRS broth

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature 37 °C

Length of Incubation 6 hrs ,depends on selection.

This transfer is briefly discussed in *Plasmid* **23**:119-125 (1990) and was taken from *FEMS Microbiol. Lett.* **44**:173-177 (1987).

Selection Method or Assay Used Antibiotic resistance (10 µg / ml erythromycin) on MRS plates

Electroporation Efficiency 2 X 10⁽⁵⁾ transformants / µg DNA

Per Cent Survival 0.03%

Name of Submitter Lori Rinckel / Grad Student

Institution Address University of Tennessee
Department of Microbiology
M409 Walters Life Science Bldg.
Knoxville, TN 37996

Survey Number

074